

The usual suspects

three generations of matter (fermions)						interactions / force carriers (bosons)	
	I	II	III				
mass	$\approx 2.16 \text{ MeV}/c^2$	$\approx 1.273 \text{ GeV}/c^2$	$\approx 172.57 \text{ GeV}/c^2$	0		$\approx 125.2 \text{ GeV}/c^2$	
charge	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{2}{3}$	0		0	
spin	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1		0	
QUARKS	u up	c charm	t top	g gluon		H higgs	
	$\approx 4.7 \text{ MeV}/c^2$ $-\frac{1}{3}$ $\frac{1}{2}$ d down	$\approx 93.5 \text{ MeV}/c^2$ $-\frac{1}{3}$ $\frac{1}{2}$ s strange	$\approx 4.183 \text{ GeV}/c^2$ $-\frac{1}{3}$ $\frac{1}{2}$ b bottom	0 0 1 γ photon			
LEPTONS	$\approx 0.511 \text{ MeV}/c^2$ -1 $\frac{1}{2}$ e electron	$\approx 105.66 \text{ MeV}/c^2$ -1 $\frac{1}{2}$ μ muon	$\approx 1.77693 \text{ GeV}/c^2$ -1 $\frac{1}{2}$ τ tau	$\approx 91.188 \text{ GeV}/c^2$ 0 1 Z Z boson			
	$< 0.8 \text{ eV}/c^2$ 0 $\frac{1}{2}$ ν_e electron neutrino	$< 0.17 \text{ MeV}/c^2$ 0 $\frac{1}{2}$ ν_μ muon neutrino	$< 18.2 \text{ MeV}/c^2$ 0 $\frac{1}{2}$ ν_τ tau neutrino	$\approx 80.3692 \text{ GeV}/c^2$ ± 1 1 W W boson			

GAUGE BOSONS
VECTOR BOSONS

SCALAR BOSONS

Baryoner

Particle name	Symbol	Quark content	Rest mass (MeV/c ²)	<i>I</i>	<i>J^P</i>	<i>Q</i> (e)	<i>S</i>	<i>C</i>	<i>B'</i>	Mean lifetime (s)	Commonly decays to
proton ^[8]	p, p^+, N^+	uud	938.272 0813(58) _[a]	$\frac{1}{2}$	$\frac{1}{2}^+$	+1	0	0	0	stable ^[b]	unobserved
neutron ^[9]	n, n^0, N^0	udd	939.565 4133(58) _[a]	$\frac{1}{2}$	$\frac{1}{2}^+$	0	0	0	0	$(8.794 \pm 0.006) \times 10^{-2}$ ^[c]	$p^+ + e^- + \bar{\nu}_e$
lambda ^[10]	Λ^0	uds	$1\,115.683 \pm 0.006$	0	$\frac{1}{2}^+$	0	-1	0	0	$(2.632 \pm 0.020) \times 10^{-10}$	$p^+ + \pi^-$ or $n^0 + \pi^0$
charmed lambda ^[11]	Λ_c^+	udc	$2\,286.46 \pm 0.14$	0	$\frac{1}{2}^+$	+1	0	+1	0	$(2.024 \pm 0.031) \times 10^{-13}$	see Λ_c^+ decay modes 
bottom lambda ^[12]	Λ_b^0	udb	$5\,619.6 \pm 0.17$	(0)	$(\frac{1}{2}^+)$	0	0	0	-1	$(1.471 \pm 0.009) \times 10^{-12}$	see Λ_b^0 decay modes 
sigma ^[13]	Σ^+	uus	$1\,189.37 \pm 0.07$	1	$\frac{1}{2}^+$	+1	-1	0	0	$(8.018 \pm 0.026) \times 10^{-11}$	$p^+ + \pi^0$ or $n^0 + \pi^+$
sigma ^[14]	Σ^0	uds	$1\,192.642 \pm 0.024$	1	$\frac{1}{2}^+$	0	-1	0	0	$(7.4 \pm 0.7) \times 10^{-20}$	$\Lambda^0 + \gamma$
sigma ^[15]	Σ^-	dds	$1\,197.449 \pm 0.030$	1	$\frac{1}{2}^+$	-1	-1	0	0	$(1.479 \pm 0.011) \times 10^{-10}$	$n^0 + \pi^-$

charmed sigma ^[16]	$\Sigma_c^{++}(2455)$	uuc	$2\,453.97 \pm 0.14$	1	$\frac{1}{2}^+$	+2	0	+1	0	$3.48^{+0.37}_{-0.16} \times 10^{-22[d]}$	$\Lambda_c^+ + \pi^+$
charmed sigma ^[16]	$\Sigma_c^+(2455)$	udc	$2\,452.9 \pm 0.4$	1	$\frac{1}{2}^+$	+1	0	+1	0	$>1.43 \times 10^{-22[d]}$	$\Lambda_c^+ + \pi^0$
charmed sigma ^[16]	$\Sigma_c^0(2455)$	ddc	$2\,453.75 \pm 0.14$	1	$\frac{1}{2}^+$	0	0	+1	0	$3.60^{+0.42}_{-0.20} \times 10^{-22[d]}$	$\Lambda_c^+ + \pi^-$
bottom sigma ^[17]	Σ_b^+	uub	$5\,810.56 \pm 0.25$	(1)	$(\frac{1}{2}^+)$	+1	0	0	-1	$(1.32 \pm 0.13) \times 10^{-22[d]}$	$\Lambda_b^0 + \pi^+$
bottom sigma [†]	Σ_b^0	udb	unknown	(1)	$(\frac{1}{2}^+)$	0	0	0	-1	unknown	unknown
bottom sigma ^[17]	Σ_b^-	ddb	$5\,815.64 \pm 0.27$	(1)	$(\frac{1}{2}^+)$	-1	0	0	-1	$(1.24 \pm 0.12) \times 10^{-22[d]}$	$\Lambda_b^0 + \pi^-$
$\xi_i^{[18]}$	Ξ^0	uss	$1\,314.86 \pm 0.20$	$\frac{1}{2}$	$\frac{1}{2}^{(+)}$	0	-2	0	0	$(2.90 \pm 0.09) \times 10^{-10}$	$\Lambda^0 + \pi^0$
$\xi_i^{[19]}$	Ξ^-	dss	$1\,321.71 \pm 0.07$	$\frac{1}{2}$	$\frac{1}{2}^{(+)}$	-1	-2	0	0	$(1.639 \pm 0.015) \times 10^{-10}$	$\Lambda^0 + \pi^-$
charmed $\xi_i^{[20]}$	Ξ_c^+	usc	$2\,467.94^{+0.17}_{-0.20}$	$(\frac{1}{2})$	$(\frac{1}{2}^+)$	+1	-1	+1	0	$(4.56 \pm 0.05) \times 10^{-13}$	see Ξ_c^+ decay modes 

charmed $\xi_i^{[21]}$	Ξ_c^0	dsc	$2\,470.90^{+0.22}_{-0.29}$	$(\frac{1}{2})$	$(\frac{1}{2}^+)$	0	-1	+1	0	$(1.53 \pm 0.06) \times 10^{-13}$	see Ξ_c^0 decay modes 
charmed ξ_i prime ^[22]	Ξ_c^+	usc	$2\,578.4 \pm 0.5$	$(\frac{1}{2})$	$(\frac{1}{2}^+)$	+1	-1	+1	0	unknown	$\Xi_c^+ + \gamma$ (seen)
charmed ξ_i prime ^[23]	Ξ_c^0	dsc	$2\,579.2 \pm 0.5$	$(\frac{1}{2})$	$(\frac{1}{2}^+)$	0	-1	+1	0	unknown	$\Xi_c^0 + \gamma$ (seen)
double charmed $\xi_i^{[24]}$	Ξ_{cc}^{++}	ucc	$3\,621.2 \pm 0.7$	$(\frac{1}{2})$	$(\frac{1}{2}^+)$	+2	0	+2	0	unknown	see Ξ_{cc}^{++} decay modes 
double charmed $\xi_i^{[e]}$	Ξ_{cc}^+	dcc	unknown	$(\frac{1}{2})$	$(\frac{1}{2}^+)$	+1	0	+2	0	unknown	unknown
bottom $\xi_i^{[25]}$ or cascade B	Ξ_b^0	usb	$5\,791.9 \pm 0.5$	$(\frac{1}{2})$	$(\frac{1}{2}^+)$	0	-1	0	-1	$(1.480 \pm 0.030) \times 10^{-12}$	see Ξ_b^0 decay modes 

bottom xi ^[25] or cascade B	Ξ_b^-	dsb	$5\,797.0 \pm 0.6$	$(\frac{1}{2})$	$(\frac{1}{2}^+)$	-1	-1	0	-1	$(1.572 \pm 0.040) \times 10^{-12}$	see Ξ_b^- decay modes 
bottom xi prime [†]	Ξ_b^0	usb	unknown	$(\frac{1}{2})$	$(\frac{1}{2}^+)$	0	-1	0	-1	unknown	unknown
bottom xi prime [†]	Ξ_b^-	dsb	unknown	$(\frac{1}{2})$	$(\frac{1}{2}^+)$	-1	-1	0	-1	unknown	unknown
double bottom xi [†]	Ξ_{bb}^0	ubb	unknown	$(\frac{1}{2})$	$(\frac{1}{2}^+)$	0	0	0	-2	unknown	unknown
double bottom xi [†]	Ξ_{bb}^-	dbb	unknown	$(\frac{1}{2})$	$(\frac{1}{2}^+)$	-1	0	0	-2	unknown	unknown
charmed bottom xi [†]	Ξ_{cb}^+	ucb	unknown	$(\frac{1}{2})$	$(\frac{1}{2}^+)$	+1	0	+1	-1	unknown	unknown
charmed bottom xi [†]	Ξ_{cb}^0	dcb	unknown	$(\frac{1}{2})$	$(\frac{1}{2}^+)$	0	0	+1	-1	unknown	unknown

charmed bottom xi prime [†]	Ξ_{cb}^+	ucb	unknown	$(\frac{1}{2})$	$(\frac{1}{2}^+)$	+1	0	+1	-1	unknown	unknown
charmed bottom xi prime [†]	Ξ_{cb}^0	dcb	unknown	$(\frac{1}{2})$	$(\frac{1}{2}^+)$	0	0	+1	-1	unknown	unknown
charmed omega ^[26]	Ω_c^0	ssc	$2\,695.2 \pm 1.7$	(0)	$(\frac{1}{2}^+)$	0	-2	+1	0	$(2.68 \pm 0.24 \pm 0.10) \times 10^{-13}$	see Ω_c^0 decay modes 
bottom omega ^[27]	Ω_b^-	ssb	$6\,046.1 \pm 1.7$	(0)	$(\frac{1}{2}^+)$	-1	-2	0	-1	$1.64^{+0.18}_{-0.17} \times 10^{-12}$	$\Omega^- + J/\psi$ (seen)
double charmed omega [†]	Ω_{cc}^+	scc	unknown	(0)	$(\frac{1}{2}^+)$	+1	-1	+2	0	unknown	unknown
charmed bottom omega [†]	Ω_{cb}^0	scb	unknown	(0)	$(\frac{1}{2}^+)$	0	-1	+1	-1	unknown	unknown
charmed bottom omega	Ω_{cb}^0	scb	unknown	(0)	$(\frac{1}{2}^+)$	0	-1	+1	-1	unknown	unknown
double bottom omega [†]	Ω_{bb}^-	sbb	unknown	(0)	$(\frac{1}{2}^+)$	-1	-1	0	-2	unknown	unknown
double charmed bottom omega [†]	Ω_{ccb}^+	ccb	unknown	(0)	$(\frac{1}{2}^+)$	+1	0	+2	-1	unknown	unknown
charmed double bottom omega [†]	Ω_{cbb}^0	cbb	unknown	(0)	$(\frac{1}{2}^+)$	0	0	+1	-2	unknown	unknown

Mesoner

Particle name	Particle symbol	Antiparticle symbol	Quark content	Rest mass (MeV/c ²)	I ^G	J ^{PC}	S	C	B'	Mean lifetime (s)	Commonly decays to (>5% of decays)
Pion ^[10]	π^+	π^-	$u\bar{d}$	139.570 18 ± 0.000 35	1 ⁻	0 ⁻	0	0	0	(2.6033 ± 0.0005) × 10 ⁻⁸	$\mu^+ + \nu_\mu$
Pion ^[11]	π^0	Self	$\frac{u\bar{u}-d\bar{d}}{\sqrt{2}}$ ^[a]	134.9766 ± 0.0006	1 ⁻	0 ⁺⁺	0	0	0	(8.52 ± 0.18) × 10 ⁻¹⁷	$\gamma + \gamma$
Eta meson ^[12]	η	Self	$\frac{u\bar{u}+d\bar{d}-2s\bar{s}}{\sqrt{6}}$ ^[a]	547.862 ± 0.018	0 ⁺	0 ⁺⁺	0	0	0	(5.02 ± 0.19) × 10 ⁻¹⁹ ^[b]	$\gamma + \gamma$ or $\pi^0 + \pi^0 + \pi^0$ or $\pi^+ + \pi^0 + \pi^-$
Eta prime meson ^[13]	$\eta'(958)$	Self	$\frac{u\bar{u}+d\bar{d}+s\bar{s}}{\sqrt{3}}$ ^[a]	957.78 ± 0.06	0 ⁺	0 ⁺⁺	0	0	0	(3.32 ± 0.15) × 10 ⁻²¹ ^[b]	$\pi^+ + \pi^- + \eta$ or ($\rho^0 + \gamma$) / ($\pi^+ + \pi^- + \gamma$) or $\pi^0 + \pi^0 + \eta$
Charmed eta meson ^[14]	$\eta_c(1S)$	Self	$c\bar{c}$	2,983.6 ± 0.7	0 ⁺	0 ⁺⁺	0	0	0	(2.04 ± 0.05) × 10 ⁻²³ ^[b]	See η_c decay modes
Bottom eta meson ^[15]	$\eta_b(1S)$	Self	$b\bar{b}$	9,398.0 ± 3.2	0 ⁺	0 ⁺⁺	0	0	0	(6.58 ± 0.15) × 10 ⁻²³ ^[b]	See η_b decay modes

Kaon ^[16]	K^+	K^-	$u\bar{s}$	493.677 ± 0.016	1/2	0 ⁻	1	0	0	(1.2380 ± 0.0021) × 10 ⁻⁸	$\mu^+ + \nu_\mu$ or $\pi^+ + \pi^0$ or $\pi^0 + e^+ + \nu_e$ or $\pi^+ + \pi^+ + \pi^-$
Kaon ^[17]	K^0	$\bar{K}^0$	$d\bar{s}$	497.614 ± 0.024	1/2	0 ⁻	1	0	0	^[c]	^[c]
K-Short ^[18]	K_S^0	Self	$\frac{d\bar{s}-s\bar{d}}{\sqrt{2}}$ ^[e]	497.614 ± 0.024 ^[d]	1/2	0 ⁻	(*)	0	0	(8.954 ± 0.004) × 10 ⁻¹¹	$\pi^+ + \pi^-$ or $\pi^0 + \pi^0$
K-Long ^[19]	K_L^0	Self	$\frac{d\bar{s}+s\bar{d}}{\sqrt{2}}$ ^[e]	497.614 ± 0.024 ^[d]	1/2	0 ⁻	(*)	0	0	(5.116 ± 0.021) × 10 ⁻⁸	$\pi^\pm + e^\mp + \nu_e$ or $\pi^\pm + \mu^\mp + \nu_\mu$ or $\pi^0 + \pi^0 + \pi^0$ or $\pi^+ + \pi^0 + \pi^-$
D meson ^[20]	D^+	D^-	$c\bar{d}$	1,869.61 ± 0.10	1/2	0 ⁻	0	+1	0	(1.040 ± 0.007) × 10 ⁻¹²	See D^+ decay modes
D meson ^[21]	D^0	$\bar{D}^0$	$c\bar{u}$	1,864.84 ± 0.07	1/2	0 ⁻	0	+1	0	(4.101 ± 0.015) × 10 ⁻¹³	See D^0 decay modes

strange D meson ^[22]	D_s^+	D_s^-	$c\bar{s}$	1,968.30 ± 0.11	0	0 ⁻	+1	+1	0	(5.00 ± 0.07) × 10 ⁻¹³	See D_s^+ decay modes
B meson ^[23]	B^+	B^-	$u\bar{b}$	5,279.26 ± 0.17	1/2	0 ⁻	0	0	+1	(1.638 ± 0.004) × 10 ⁻¹²	See B^+ decay modes
B meson ^[24]	B^0	$\bar{B}^0$	$d\bar{b}$	5,279.58 ± 0.17	1/2	0 ⁻	0	0	+1	(1.519 ± 0.009) × 10 ⁻¹²	See B^0 decay modes
Strange B meson ^[25]	B_s^0	$\bar{B}_s^0$	$s\bar{b}$	5,366.77 ± 0.24	0	0 ⁻	-1	0	+1	(1.512 ± 0.007) × 10 ⁻¹²	See B_s^0 decay modes
Charmed B meson ^[26]	B_c^+	B_c^-	$c\bar{b}$	6,275.6 ± 1.1	0	0 ⁻	0	+1	+1	(4.52 ± 0.33) × 10 ⁻¹³	See B_c^+ decay modes